

Networks and Systems Management



DNS
Filipe Sousa
FEUP – GRS/M.EIC

Introduction to DNS

What is DNS?

- Translates domain names to IP addresses
- Essential for internet navigation
- Hierarchical and distributed system

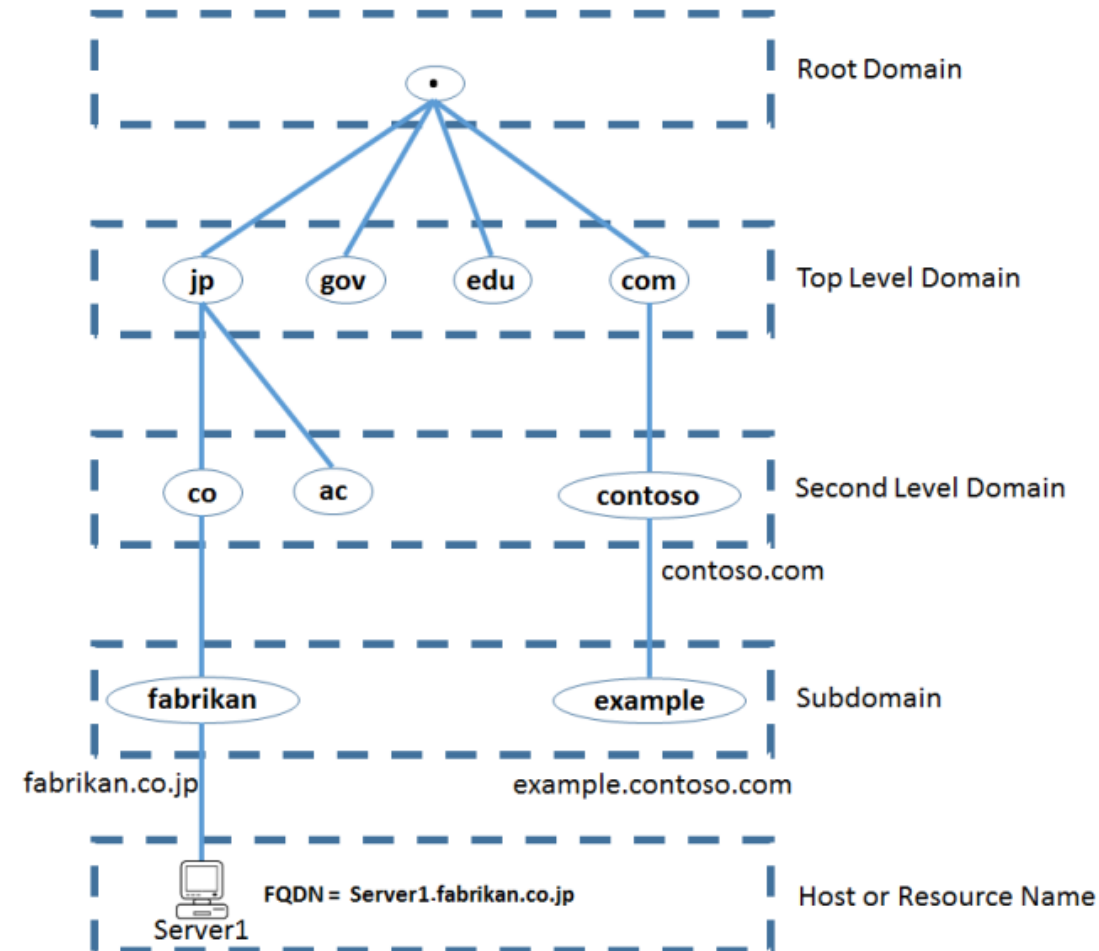
Why is it important?

- People use domain names to access web sites
- IP is used to route information in internet
- DNS maps IP addresses and domain name



DNS Hierarchy

- Structure:
 - Root Level: (".") The starting point of all DNS lookups
 - Top-Level Domains (TLDs): .com, .org, .net, .gov, etc.
 - Second-Level Domains: Example: google.com, wikipedia.org
 - Subdomains: Example: mail.google.com, en.wikipedia.org
- Visualization:
 - Tree-like structure with delegation at each level



DNS Architecture



Components:

Root Name Servers

Top-Level Domain (TLD)
Servers

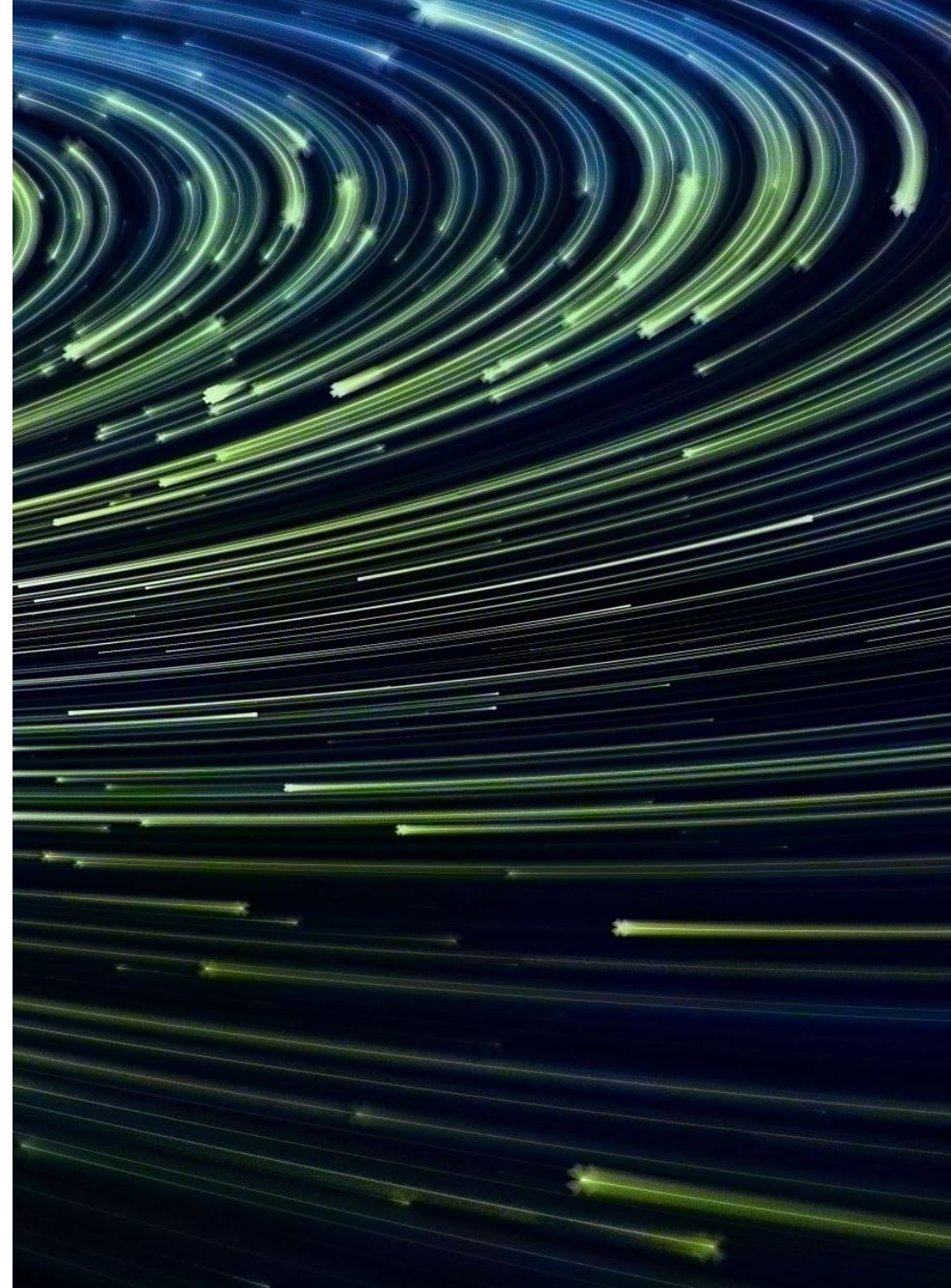
Authoritative Name Servers

Recursive Resolvers



Hierarchy & Flow:

How a DNS query traverses
these components



Root Authoritative Servers

- A DNS zone is a portion of the DNS namespace that is managed by a specific organization or administrator
- The DNS root zone contains all domain names
- Root servers respond to TLD requests

List of Root Servers

HOSTNAME	IP ADDRESSES	OPERATOR
a.root-servers.net	198.41.0.4, 2001:503:ba3e::2:30	Verisign, Inc.
b.root-servers.net	170.247.170.2, 2801:1b8:10::b	University of Southern California, Information Sciences Institute
c.root-servers.net	192.33.4.12, 2001:500:2::c	Cogent Communications
d.root-servers.net	199.7.91.13, 2001:500:2d::d	University of Maryland
e.root-servers.net	192.203.230.10, 2001:500:a8::e	NASA (Ames Research Center)
f.root-servers.net	192.5.5.241, 2001:500:2f::f	Internet Systems Consortium, Inc.
g.root-servers.net	192.112.36.4, 2001:500:12::d0d	US Department of Defense (NIC)
h.root-servers.net	198.97.190.53, 2001:500:1::53	US Army (Research Lab)
i.root-servers.net	192.36.148.17, 2001:7fe::53	Netnod
j.root-servers.net	192.58.128.30, 2001:503:c27::2:30	Verisign, Inc.
k.root-servers.net	193.0.14.129, 2001:7fd::1	RIPE NCC
l.root-servers.net	199.7.83.42, 2001:500:9f::42	ICANN
m.root-servers.net	202.12.27.33, 2001:dc3::35	WIDE Project

<https://www.iana.org/domains/root/servers>

Non-root Authoritative Servers

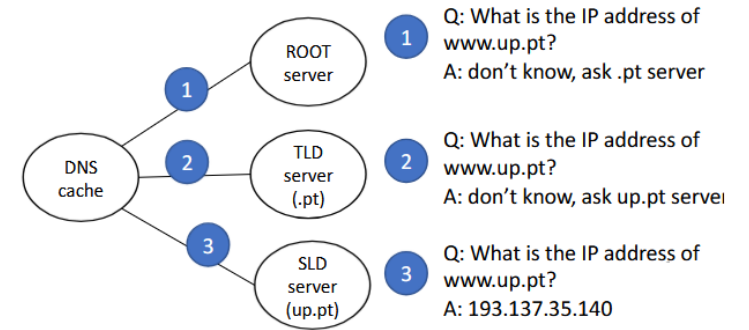
- Top Level Domains
 - gTLD – Generic Top-Level Domains
 - Original: .com, .org, .net, .int, .edu, .gov, .mil,
 - 1502 new TLDs as of April 2021, internacionalised
 - https://en.wikipedia.org/wiki/List_of_Internet_to_p-level_domains
 - ccTLD – Country Code Top-Level Domains
 - 316 ccTLDs as of June 2020, including internationalized
 - https://en.wikipedia.org/wiki/Country_code_top-level_domain

Non-root Authoritative Servers

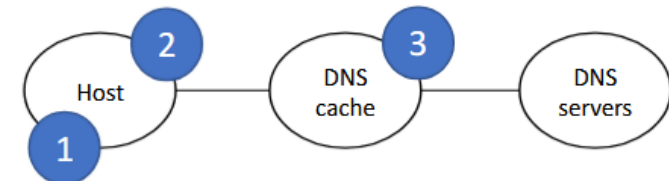
- Second-level domains
 - Each TLD has a TLD manager organization responsible for assigning SLDs under that TLD
 - The same happens down the hierarchy until the final host or resource name
- Masters
 - stores the definitive versions of all records
 - identified in start-of-authority (SOA) resource record
- Slaves
 - automatic updating mechanism to maintain an identical copy of the primary server's database for a zone
 - DNS zone transfer, AXFR,
https://en.wikipedia.org/wiki/DNS_zone_transfer

Architecture and Protocol

- Local
 - /etc/hosts, resolv.conf, DHCP
 - Non-recursive, recursive, iterative
- DNS Request and reply messages
 - QR, AA, NAME, TYPE
 - Over TCP or UDP ports 53; DoT, DoH, DNS over TOR, ...
- Servers
 - Local/remote DNS server/cache/resolver
 - Root name server, TLD, SLD
 - Caching
 - Split server



- 1 If I know the IP address, use it.
- 2 If I don't, ask the local server/cache.
- 3 If not in cache, ask other DNS servers.



Resolver and Cache

- Recursive

vs.

- Iterative

- Referral

- NS: "can't find this name, try this NS"
- (2) and (4) in red in the figure

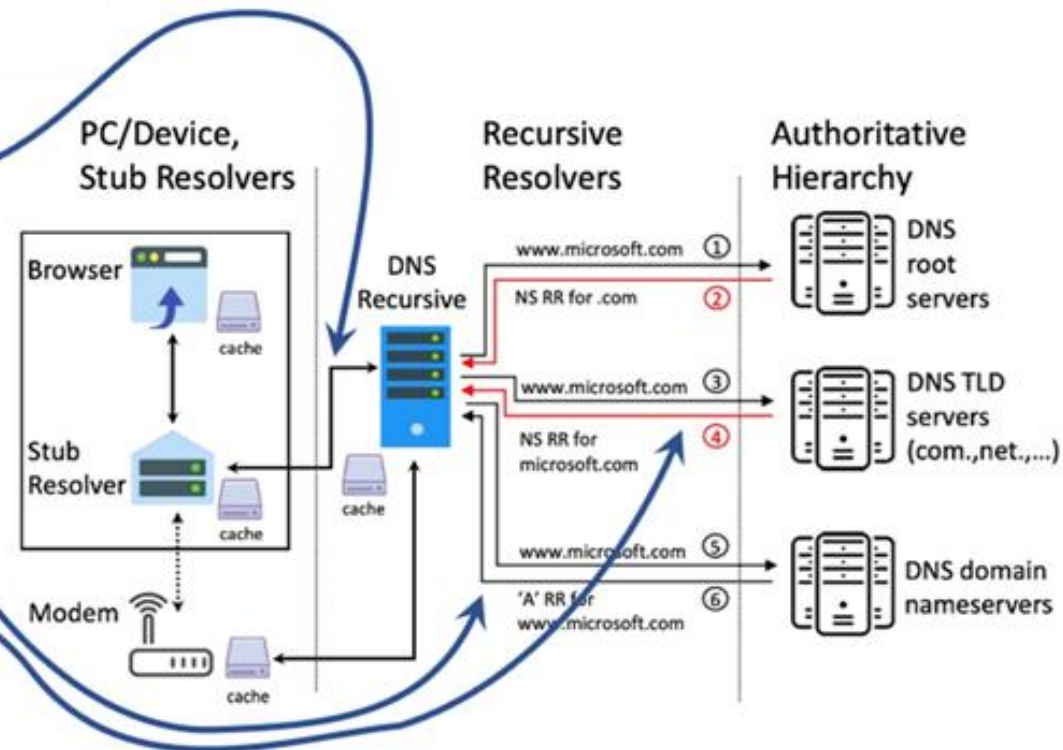
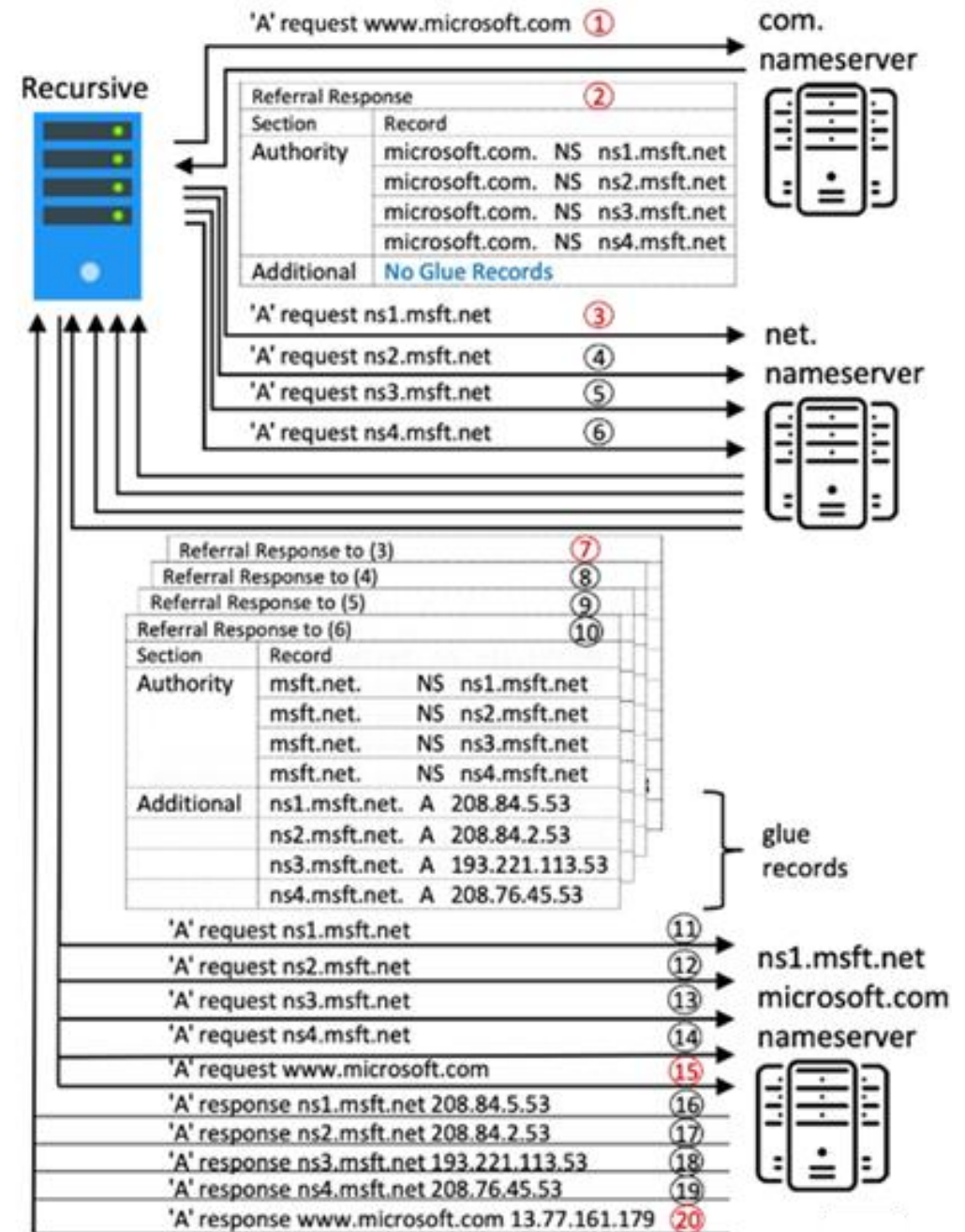


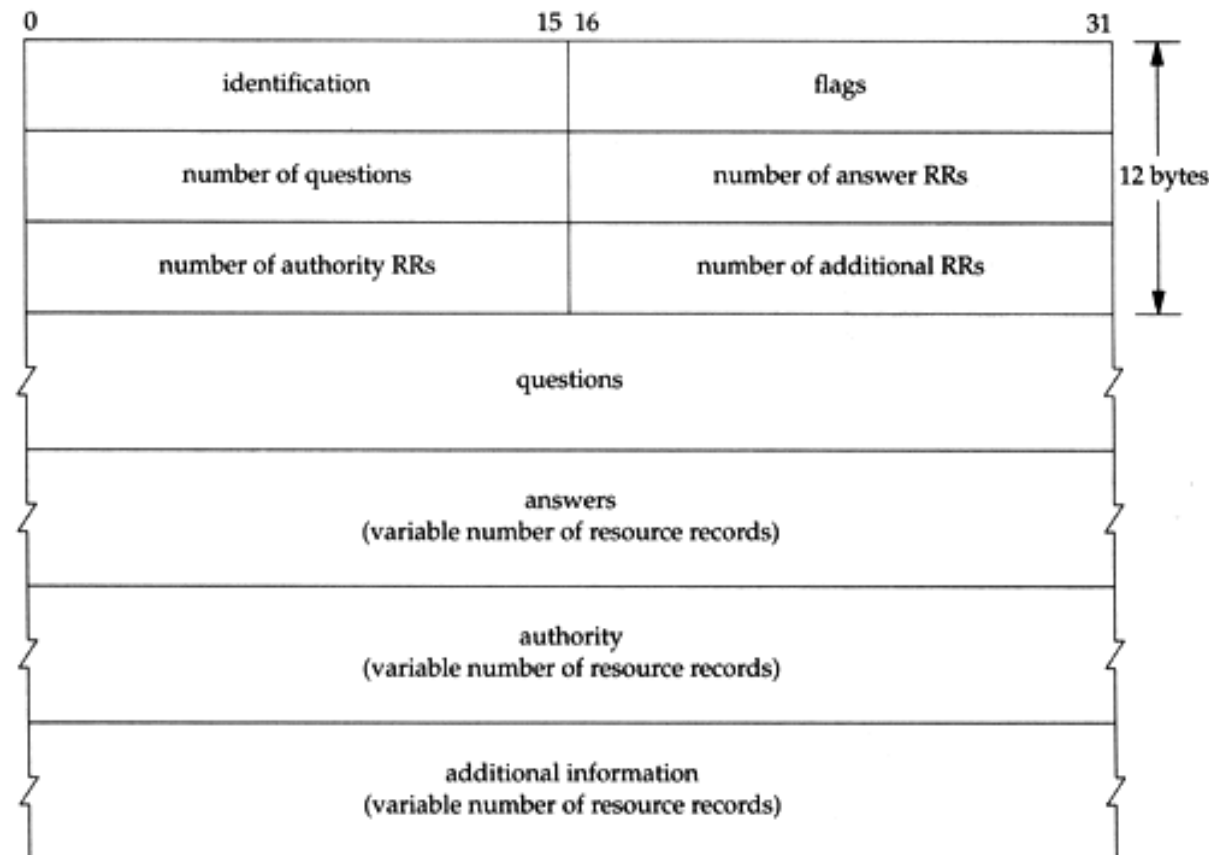
Figure 1: The resolution process, in theory, for the domain `www.microsoft.com`. The red steps represent NS referral responses.

Redundancy overhead



DNS Message Structure

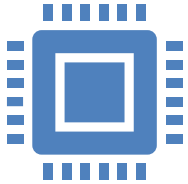
- **Header Fields:**
 - Query ID, Flags, Question Count, Answer Count, Authority Count, Additional Count
- **Sections:**
 - Query Section
 - Answer Section
 - Authority Section
 - Additional Section



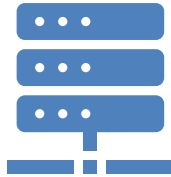
Common DNS Records

- **A Record:** Maps domain to IPv4 address
- **AAAA Record:** Maps domain to IPv6 address
- **CNAME Record:** Alias for another domain
- **MX Record:** Mail exchange information
- **NS Record:** Specifies authoritative name servers
- **TXT Record:** Contains text information (e.g., SPF for email security)

DNS Security Threats



Cache Poisoning (DNS Spoofing): Malicious redirection



DDoS Attacks: Overloading DNS servers



Man-in-the-Middle (MITM) Attacks: Intercepting DNS queries



DNS Tunneling: Exfiltrating data via DNS queries

Securing DNS



DNSSEC (DNS Security Extensions):

Adds cryptographic signatures to DNS responses



DoH (DNS over HTTPS) & DoT (DNS over TLS):

Encrypts DNS queries



Implementing Firewalls & Rate Limiting:

Prevents abuse and attacks



Regular DNS Auditing & Monitoring:

Detects anomalies and unauthorized changes

Best Practices for DNS Security

- Use secure, reputable DNS resolvers
- Enable DNSSEC for domain integrity
- Enforce DoH/DoT where applicable
- Monitor DNS logs for unusual activity
- Limit zone transfers and restrict recursive queries

Key Takeaways

DNS is a hierarchical system that enables internet functionality.

Understanding its architecture is crucial for security and performance.

DNS security mechanisms like DNSSEC, DoH, and DoT help mitigate threats.

Regular auditing and best practices enhance DNS resilience.

Further Reading & References

- RFC 1035 - Domain Names Implementation and Specification
- RFC 4033 - DNS Security Introduction and Requirements
- Cloudflare: "What is DNSSEC?"
- ICANN: "DNS Security Best Practices"
- Google Public DNS Documentation

